

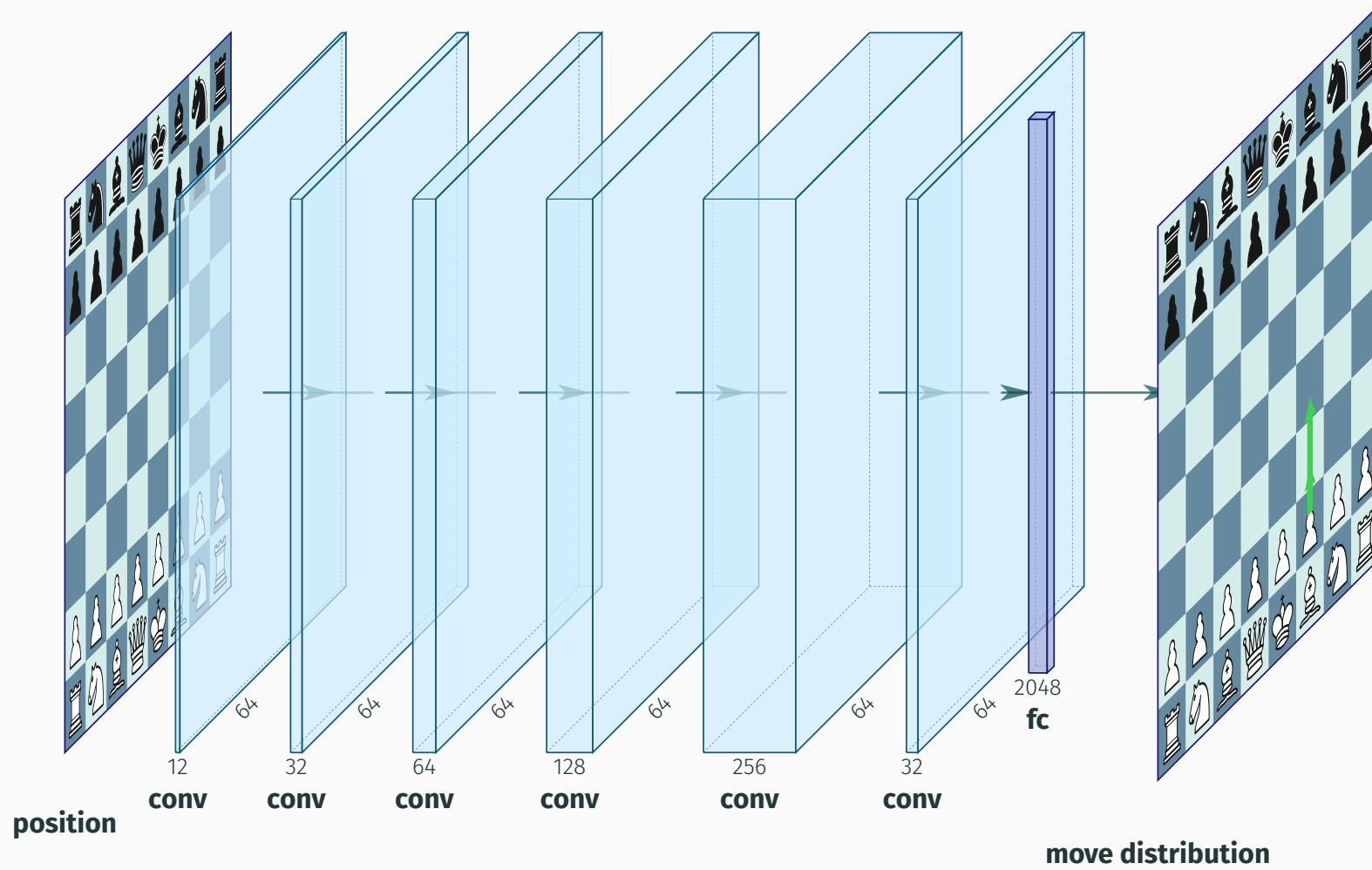
# Exploring Interaction Between Behavior Cloning and Reward Shaping in Chess

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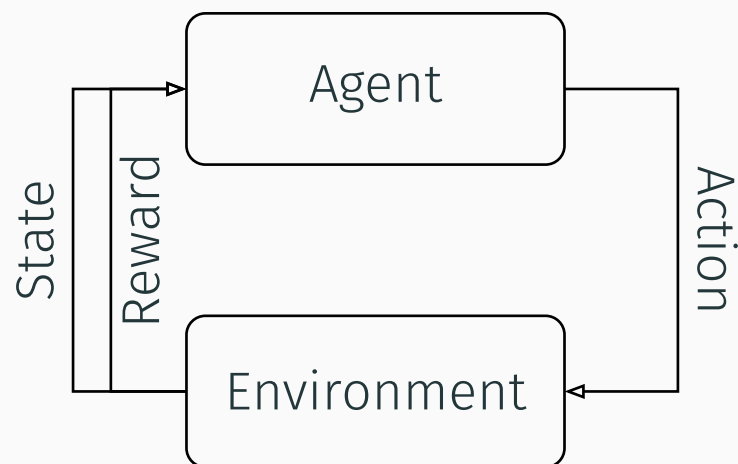
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# Introduction



# Reinforcement Learning

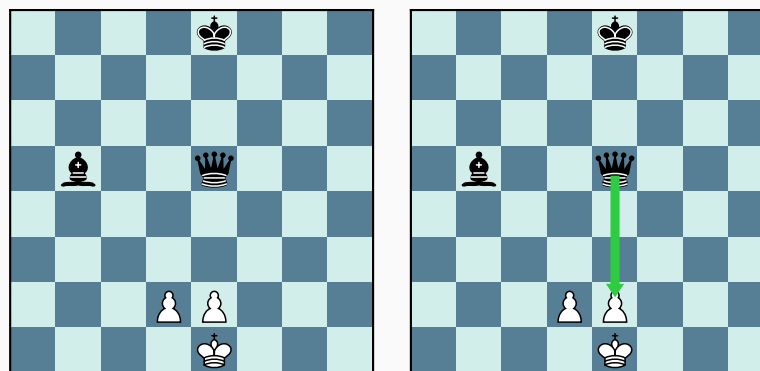
## Reinforcement Learning



## Problems

- Sparse rewards → poor performance
- Reward Shaping → difficult [1]:
  - Missaligned rewards
  - Reward hacking
  - Prioritizing wrong rewards

## Imitation Learning: Behavior Cloning



- Supervised learning
- Agent learns expert demonstrations
- e.g. puzzles, grandmaster games

## Problems

- Demonstration sample bias → poor performance [2]

## **Combine both approaches**

1. Pretrain: Behavior Cloning
2. Finetune: Reward Based Reinforcement Learning

## **Can both approaches stabilize each other?**

- Behavior Cloning: Stabilize RL with sparse rewards
- Reinforcement Learning: Compensates for data bias

## **Lets find out!**

- Train different models:
  - Architecture: Fully connected, cnn, residual block
  - Pretraining Data: Heavily biased → less biased
  - Rewards Sets: Sparse → Dense

# Architecture



# Architecture

## FCNN

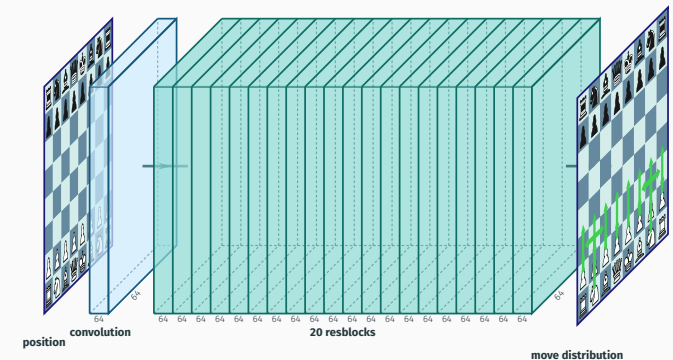
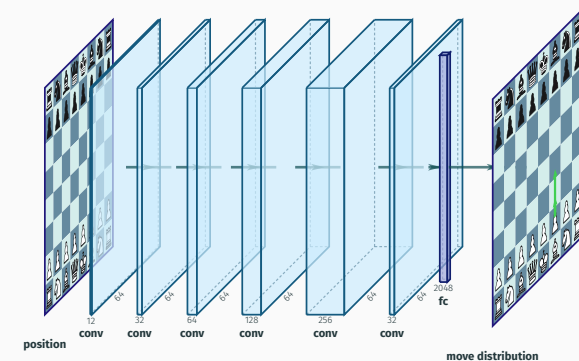
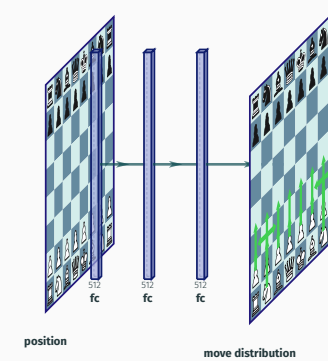
- 3 hidden layers: 512 neurons

## CNN

- increasing number of channels:
- $12 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 32$

## ResNet

- 20 blocks with 64 channels

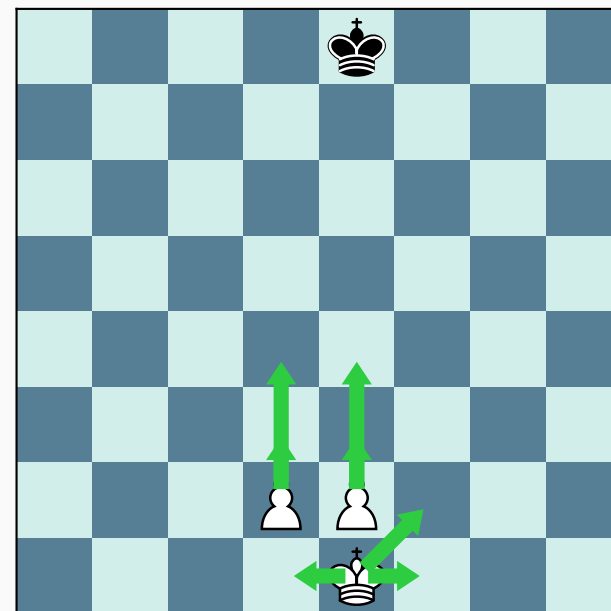


## Input

- tensor shape =  $(8, 8, 12)$ :
  - $8 \times 8$  squares
  - 12 pieces: {White, Black}  $\times$  {Pawn, Rook, Knight, Bishop, Queen, King}

## Output

- Move distribution: from square  $\rightarrow$  to square
- **But:** Not all moves are legal  $\rightarrow$  mask illegal moves

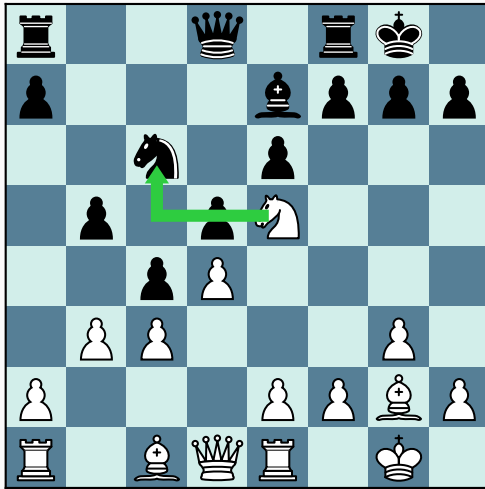


# Imitation Learning

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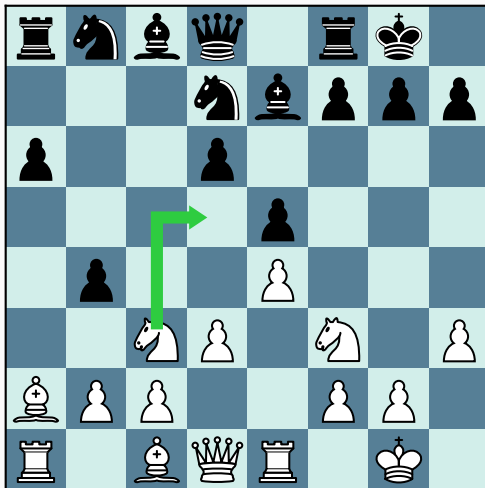


## Strategy vs Tactics



### Tactics

- Short term moves
  - winning a piece
  - checkmate
  - etc.



### Strategy

- Long term move:
  - Good piece positioning
  - Pressure on opponent King
  - Control of center
  - etc.

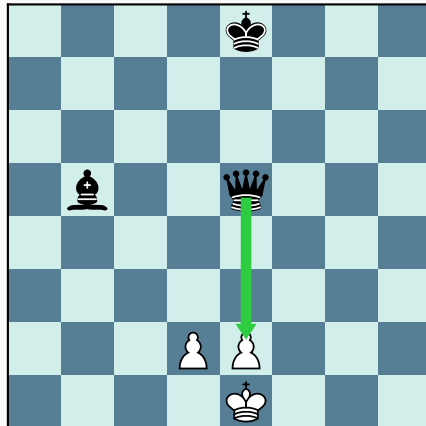
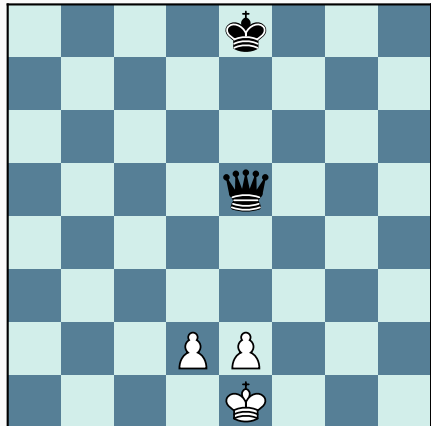
# Imitation Learning: Behavior Cloning

## Lichess Puzzles [3]

- 6 million positions
- tactical positions

## Bias

- no strategic positions
- no positions without tactic

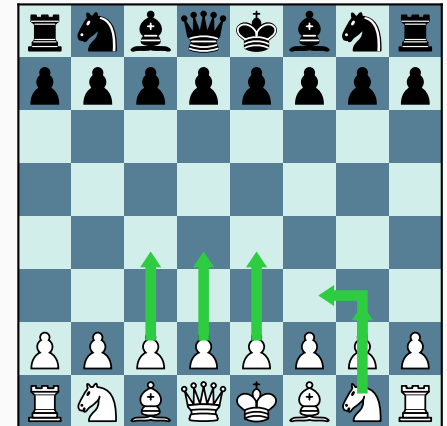
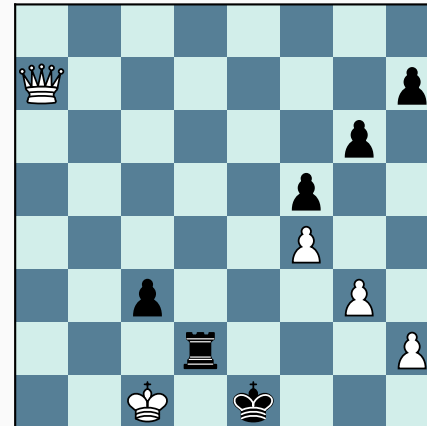


## GM games [4]

- chess.com master games
- balance tactics/strategy

## Bias & Problems

- high level play
- many resignations
- multiple moves possible



# Reinforcement Learning

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- **Goal:** Directly optimize a stochastic policy  $\pi_{\theta}(a \mid s)$  by maximizing the expected return

$$J(\theta) = E_{\tau \sim \pi_{\theta}} \left[ \sum_{t=0}^T R_t \right].$$

- **Core idea:** Increase the probability of actions that lead to high returns

$$\nabla_{\theta} J(\pi_{\theta}) = E_{\tau \sim \pi_{\theta}} \left[ \sum_{t=0}^T \nabla \log \pi_{\theta}(a_t \mid s_t) G_t \right].$$

- **Algorithm:**
  - Sample trajectories using the current policy  $\pi_{\theta}$
  - Compute the return  $G_t$  for each time step
  - Update parameters via gradient ascent

## Reward sets

0. **Sparse:** rewards only for true goal
  - reward for win/loss/draw
1. **Medium:** more rewards, but might misalign with goals
  - control center, material win, king safety
2. **Dense:** highest risk of misalignment
  - control outer center, control center, castling, promoting, blunder prevention, material, king safety, give check, win

## Training

- Architecture: {FCNN, CNN, ResNet}
- Pretraining: {Puzzles, Masters}  $\times$  {1 Epoch, 10 Epochs}, {Untrained}
- Reward Set: {R\_0, R\_1, R\_2}
  - 1000 batches with 16 games each (REINFORCE)

→ 57 models

# Results

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# Behavior Cloning: Training Accuracy

| dataset       | epochs | fc    | cnn   | resnet |
|---------------|--------|-------|-------|--------|
| puzzle        | 1      | 62.0% | 69.9% | 74.6%  |
|               | 10     | 65.8% | 73.3% | 81.7%  |
| grand masters | 1      | 31.0% | 38.5% | 41.5%  |
|               | 10     | 31.8% | 39.6% | 43.7%  |

- Puzzle better generalizing
- **But** master games: different labels, same position



# Behavior Cloning

## Evaluation

- All models plays 1000 games against all other model (same architecture)
- Normalized tournament score (win 1 point, draw 0.5 points, loss 0 points)

## BC model with RL

| dataset       | epochs | fc    | cnn   | resnet |
|---------------|--------|-------|-------|--------|
| none          |        | 0.310 | 0.251 | 0.405  |
| puzzle        | 1      | 0.615 | 0.469 | 0.515  |
|               | 10     | 0.562 | 0.538 | 0.533  |
| grand masters | 1      | 0.472 | 0.556 | 0.510  |
|               | 10     | 0.494 | 0.624 | 0.513  |

- Models improve significantly with pretraining

# Reward Shaping

## RL models with Pretraining

| rewards | fc    | cnn   | resnet |
|---------|-------|-------|--------|
| none    | 0.461 | 0.419 | 0.855  |
| r_0     | 0.520 | 0.547 | 0.404  |
| r_1     | 0.504 | 0.532 | 0.405  |
| r_2     | 0.507 | 0.486 | 0.407  |

## RL models without Pretraining

| rewards | fc    | cnn   | resnet |
|---------|-------|-------|--------|
| r_0     | 0.289 | 0.266 | 0.408  |
| r_1     | 0.307 | 0.235 | 0.408  |
| r_2     | 0.335 | 0.251 | 0.399  |

- Depending on the model RL hurts performance
- Sparse rewards generally better

# Conclusion

- We could not find any training strategy across architectures
- Overall denser rewards hurt model performance

## **Possible Improvements**

- **Data Prep:** Unify data in Masters games
- **Architecture:** Hyperparameter tuning on pretrain accuracy
- **BC:** Mix of master games and Puzzles
- **RL:** Longer training
- **RL:** Iterative reward shaping

# How does it really play?

Well.... <https://lichess.org/cuRJkeJ1#0>

# References

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  - [2] “Imitation Learning, Stanford University.” [Online]. Available: [https://web.stanford.edu/class/cs237b/pdfs/lecture/lecture\\_10111213.pdf](https://web.stanford.edu/class/cs237b/pdfs/lecture/lecture_10111213.pdf)
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  - [4] “All GM Chess Games on Chess.com.” [Online]. Available: <https://www.kaggle.com/datasets/dimitrioskourtikakis/gm-games-chesscom>